The underlying goal of this chapter is to provide the necessary theory and mathematical framework to properly incorporate transaction costs into the portfolio optimization. This chapter builds on the findings from "Investing and Trading Consistency: Does VWAP Compromise the Stock Selection Process?" Kissell and Malamut in Journal of Trading, Fall, 2007. We expand on those concepts by providing the necessary mathematical models and quantitative framework. The process is reinforced with graphical illustrations[1].

PORTFOLIO OPTIMIZATION AND CONSTRAINTS

Many quantitative portfolio managers construct their preferred investment portfolios following the techniques introduced by Markowitz (1952) and Sharpe (1964). But often during the optimization process these managers will incorporate certain constraints into the process. These constraints are used by managers for many different reasons and in many different ways. For example, to reflect certain views or needs, to provide an additional layer of safety, or to ensure the results provide more realistic expectations. Some of the more common reasons for incorporating constraints into the portfolio optimization are:

- Fund Mandates
- Maximum Number of Names
- Reflect Future Views
- Risk Management
- Transaction Cost Management

Fund Mandates. Many funds have specified guidelines for their portfolios. Optimization constraints are used to ensure the resulting optimal mix adheres to these strategies. For example, these may specify a predetermined asset allocation process that requires certain percentages to be invested across stocks, bonds, cash, etc. These mandates may also define a predetermined max exposure to a risk factor or a sector. And some index funds may not be allowed to have their tracking error to a benchmark exceed a certain level regardless of alpha expectations.

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